

Onkareshwar Sharma

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EDUCATION

- International Institute of Information Technology (IIIT) Naya Raipur** Naya Raipur, India
B.Tech in Computer Science & Engineering (CGPA: 7.84) *July 2024 – May 2028*

SKILLS

Languages: Python, C++, C, SQL, JavaScript, TypeScript
Backend & Frameworks: Django, Flask, Node.js, REST API Design, Server-Sent Events (SSE), OCR
Databases: PostgreSQL, MongoDB, MySQL, Redis
Web Development: React.js, HTML, CSS, Postman
Cloud & DevOps: AWS, Render, Vercel Serverless, Git, GitHub, Linux, VS Code
Security & Systems: SSRF Protection, Rate Limiting, DNS/TLS, JWT/API Auth

OPEN SOURCE

- GrowthBook** | *TypeScript, Node.js, REST API, React (Remote)* Aug 2026
 - PR #1:** Engineered and merged a public **REST API** endpoint (`POST /experiments/:id/comment`) enabling programmatic comment posting, reusing internal backend logic to ensure consistent permission checks, data models, and validation.
 - PR #2:** Resolved a client-side **pagination bug** within the Saved Groups interface by synchronizing search filters with **pagination state**, restoring accurate data retrieval and search functionality across multi-page datasets.
- Onyx** | *Python, GitLab API, Backend Development (Remote)* Jul 2026
 - Designed and implemented branch-specific syncing for Onyx's **GitLab connector**, extending the backend to fetch code from a configurable branch instead of only the repository's default.
 - Refactored connector logic across GitlabConnector to thread a new **branch parameter** through two distinct **GitLab API** calls (file listing and file content fetch), preventing mismatched file states.
 - Added robust **error handling** by catching **GitlabGetError**/**GitlabHttpError** and raising descriptive **ValueError** exceptions instead of raw tracebacks.
 - Wrote and validated **regression tests** against a live GitLab repo, covering branch override, default-branch fallback, and invalid-branch paths, while maintaining full **backward compatibility** with prior default behavior.
- Tenstorrent tt-metal** | *C/C++, Kernel Programming, System Software (Remote)* Jun 2026
 - Merged contribution modernizing **kernel-level C/C++** type declarations across Quasar and Wormhole components; collaborated with maintainers through issue discussions and code review.

PROJECTS

- CSC Sarthi – AI-Powered Pre-Submission Assistant** | *Python, Django, Scikit-learn, OCR, Render* [\[Live\]](#) | [\[GitHub\]](#)
 - Architected an **AI-powered pre-submission validation platform** for **30+ government services**, performing real-time form validation and rejection-risk prediction.
 - Developed a **machine learning risk prediction engine** using service schema features, achieving **~87% validation accuracy** with **field-level explainability**.
 - Constructed an **OCR-based document verification pipeline** to extract Name, ID, and DOB, detecting **3+ types of mismatches** by cross-validating documents against form inputs.
 - Created a **WhatsApp chatbot workflow** for remote document pre-verification and deployed a **Django analytics dashboard** visualizing the **top 5 rejection causes** across **30+ districts**.
- Website Security Scanner API** | *Node.js, Vercel Serverless, REST API, DNS/TLS* [\[Live\]](#) | [\[GitHub\]](#)
 - Programmed and **shipped a REST API** running **10 parallel security checks** (HTTP security headers, TLS/SSL certificate validation, DNS-based email spoofing protection via SPF/DMARC, CORS misconfiguration, cookie security, exposed credential files, CMS fingerprinting) on any given URL, returning a scored **JSON risk report**.
 - Published** the API on the RapidAPI marketplace with **tiered subscription** pricing, growing to **264+ users**.
 - Implemented **SSRF protections** (blocking private IP ranges, localhost, and cloud metadata endpoints), per-IP rate limiting, and honest failure reporting to prevent false-positive results on unreachable targets.
 - Launched a companion public-facing **web demo** enabling non-technical users to run live scans directly in-browser without an API key, serving as an organic funnel into the paid API.
- LiteLLM – LLM Gateway & Access Management Platform** | *React, Node.js, PostgreSQL, MongoDB, Redis, Stripe* [\[GitHub\]](#)
 - Devised a Redis-based hybrid rate limiter using TTL windows and a hidden **~100K token ceiling** to prevent API billing abuse in a multi-subscription LLM gateway

ACHIEVEMENTS

- Winner, Hackathon** Mar 2026
E-Summit 2026 (Organized by E-Cell, IIIT NR)
 - Secured first place by pitching a civic-tech automation tool to an expert panel, showcasing live document parsing and compliance-check capabilities.
- Winner, Case Study Competition** Feb 2026
Xcelarate 2026 (Organized by Money Matters, IIIT NR)